

Covariance whitening gives the missing noncollapsing Polchinski flow

Automated mathematical discovery run `fa_banach_001`, lane 11

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Status. Likely valid full answer to the rescaling question in Section 3 of arXiv:2208.08186. The answer is matrix-valued, works for every finite-dimensional positive-definite covariance decomposition, and reduces to the known Langevin/Ornstein–Uhlenbeck construction for the heat-kernel decomposition.

Source question

Serres considers

$$V_t = -\log(\gamma_{C_t} * e^{-V_0}), \quad \nu_t = e^{V_\infty(0) - V_t(x)} \gamma_{C_\infty - C_t},$$

where $0 = C_0 \nearrow C_\infty$, and observes that $\nu_t \rightarrow \delta_0$. The paper asks for a rescaling of the Polchinski semigroup having a non-Dirac ergodic law, so that a transport to ν_0 can be constructed. It further asks that the rescaling remove the correction generated by the Gaussian Hessian $(C_\infty - C_t)^{-1}$ in its quantity α'_t .

$$(8) \quad C_P(\mu_s) \leq e^{\int_s^t k'_s ds} C_P(\mu_t).$$

We recognize the same result as (3) in Section 2. This transport map approach also allows to derive such results for higher order eigenvalues.

In the case of the Polchinski flow $(\nu_t)_t$ defined in Section 4.1, the distributions ν_t interpolate between ν_0 given in (9) and $\nu_\infty = \delta_0$ the Dirac delta function. Therefore, this dynamic cannot give a transport map, since ν_0 cannot be realized as the pushforward of the Dirac delta function. However, a natural question is to find a rescaling of the Polchinski semigroup such that the ergodic distribution ν_∞ is no longer the Dirac delta function, so a transport map between ν_0 and ν_∞ could be constructed. The theorem 2, linked to the monotonicity result (8) valid when the transport maps exist and are Lipschitz, strongly suggests that such a rescaling is possible. Moreover, this rescaling must remove the term $e^{(\alpha_t - \alpha_s)}$ in (17), and thus the largest eigenvalue α'_t of the matrix $(C'_t)^{\frac{1}{2}} (\nabla^2 V_t + (C_\infty - C_t)^{-1}) (C'_t)^{\frac{1}{2}}$ should characterize the order of magnitude of the rescaling. However, the problem of finding exactly such a rescaling is still open.

Write $C = C_\infty$ and $\Sigma_t = C - C_t$. We assume $C > 0$ and $\Sigma_t > 0$ at every finite time; singular intermediate covariances follow on their common supports or by approximation. Put

$$Z = \int e^{-V_0} d\gamma_C, \quad \rho_0 = Z^{-1} e^{-V_0}.$$

Thus $e^{V_\infty(0)} = Z^{-1}$ in the source normalization.

Proof intuition

The part of ν_t that collapses is completely explicit: its Gaussian covariance is Σ_t . Choose a linear map M_t carrying the fixed Gaussian γ_C to γ_{Σ_t} , and observe the flow in the blown-up coordinate $y = M_t^{-1}x$. The convolution increments then become Gaussian transitions

$$y \mapsto M_s^{-1}M_t y + \text{noise}$$

which preserve the same γ_C at every step. A single covariance identity proves both invariance and the two-parameter semigroup law. In these coordinates the varying quadratic Gaussian potential is absorbed into the fixed reference measure; only the interaction density remains in the transport velocity. This is why the Gaussian part of α'_t vanishes.

The exact rescaling

Choose any differentiable $M_t \in GL(d)$ such that

$$M_t C M_t^\top = \Sigma_t, \quad M_0 = I. \quad (1)$$

For example, $M_t = \Sigma_t^{1/2} C^{-1/2}$ works, with no commutativity assumption. Define

$$\mu_t = (M_t^{-1})_\# \nu_t, \quad \rho_t(y) = \frac{d\mu_t}{d\gamma_C}(y) = Z^{-1} e^{-V_t(M_t y)}. \quad (2)$$

For $0 \leq s \leq t$, set

$$K_{s,t} = M_s^{-1} M_t, \quad N_{s,t} = M_s^{-1} (C_t - C_s) M_s^{-\top}. \quad (3)$$

The key identity is

$$N_{s,t} = C - K_{s,t} C K_{s,t}^\top \succeq 0. \quad (4)$$

Indeed, conjugating $\Sigma_s - \Sigma_t = C_t - C_s$ by M_s^{-1} and using (1) gives (4). Let

$$Q_{s,t} f(y) = \int f(K_{s,t} y + z) d\gamma_{N_{s,t}}(z). \quad (5)$$

Theorem 1 (Noncollapsing conjugacy). *The operators $Q_{s,t}$ are Gaussian Markov kernels preserving γ_C , satisfy the two-parameter evolution law, and obey*

$$\rho_t = Q_{s,t} \rho_s. \quad (6)$$

Consequently, (μ_t) is an inhomogeneous Ornstein–Uhlenbeck evolution with the fixed invariant law γ_C . Moreover,

$$\|\mu_t - \gamma_C\|_{\text{TV}} \longrightarrow 0. \quad (7)$$

Thus (2) is a rescaling of the Polchinski flow whose ergodic distribution is the nondegenerate Gaussian γ_C .

Proof. If $Y \sim \gamma_C$ and $Z \sim \gamma_{N_{s,t}}$ are independent, then (4) gives $K_{s,t} Y + Z \sim \gamma_C$; hence $Q_{s,t}$ preserves γ_C . The convolution relation

$$e^{-V_t(x)} = \int e^{-V_s(x+\xi)} d\gamma_{C_t - C_s}(\xi)$$

evaluated at $x = M_t y$ and changed by $\xi = M_s z$ gives

$$\rho_t(y) = \int \rho_s(M_s^{-1} M_t y + z) d\gamma_{N_{s,t}}(z),$$

which is (6).

For $r \leq s \leq t$, $K_{r,s}K_{s,t} = K_{r,t}$. The covariance of the two successive noises is

$$N_{r,s} + K_{r,s}N_{s,t}K_{r,s}^\top = C - K_{r,t}CK_{r,t}^\top = N_{r,t}.$$

This proves the evolution law (with the corresponding composition order).

At $s = 0$, $K_{0,t} = M_t \rightarrow 0$ and $N_{0,t} = C_t \rightarrow C$, while

$$\rho_t(y) = \int \rho_0(M_t y + z) d\gamma_{C_t}(z). \quad (8)$$

If ρ_0 is bounded and continuous, (8) converges pointwise to $\int \rho_0 d\gamma_C = 1$ and hence in $L^1(\gamma_C)$ by dominated convergence. For general $\rho_0 \in L^1(\gamma_C)$, approximate it in L^1 by bounded continuous functions. Markov invariance implies

$$\|Q_{0,t}f\|_{L^1(\gamma_C)} \leq \|f\|_{L^1(\gamma_C)},$$

so the approximation is uniform in t . Therefore $\rho_t \rightarrow 1$ in $L^1(\gamma_C)$, which is (7). $\square$

The literal conjugate of the source's Polchinski semigroup is also explicit. With

$$P_{s,t}f(x) = e^{V_t(x)} \int f(x + \xi) e^{-V_s(x+\xi)} d\gamma_{C_t-C_s}(\xi),$$

define

$$\tilde{P}_{s,t}g(y) = P_{s,t}(g \circ M_s^{-1})(M_t y).$$

The same change of variables gives the Doob formula

$$\tilde{P}_{s,t}g = \frac{Q_{s,t}(g\rho_s)}{\rho_t}. \quad (9)$$

Thus the result is a rescaling of the full semigroup, not only of its one-time marginal measures.

The transport velocity and disappearance of α

Set

$$G_t = M_t^{-1}\dot{M}_t, \quad 2D_t = M_t^{-1}\dot{C}_tM_t^{-\top}, \quad H_t = G_t + D_tC^{-1}. \quad (10)$$

Differentiating (1) gives

$$G_tC + CG_t^\top = -2D_t, \quad H_tC + CH_t^\top = 0. \quad (11)$$

The infinitesimal density generator in (5) is

$$\mathcal{A}_t f = D_t : \text{Hess } f + \langle G_t y, \nabla f \rangle = D_t : \text{Hess } f - \langle D_tC^{-1}y, \nabla f \rangle + \langle H_t y, \nabla f \rangle. \quad (12)$$

The first two terms are reversible for γ_C ; the last is a γ_C -preserving rotation. If p_t denotes the Lebesgue density of μ_t , then (12) is exactly

$$\partial_t p_t + \nabla \cdot (v_t p_t) = 0, \quad v_t(y) = -D_t \nabla \log \rho_t(y) - H_t y. \quad (13)$$

Indeed, $H_t^\top C^{-1} + C^{-1}H_t = 0$ and $\text{Tr } H_t = 0$, so $\nabla \cdot (H_t y p_t) = \gamma_C \langle H_t y, \nabla \rho_t \rangle$.

Equation (13) supplies the source's requested Lagrangian construction. Under the standard local-Lipschitz and nonexplosion assumptions, its integral curves transport μ_s to μ_t . The inverse maps transport μ_t to

$\mu_0 = \nu_0$; whenever these inverses converge under the usual integrable Hessian bounds, (7) shows that their limit transports γ_C to ν_0 .

The cancellation mechanism is now visible. In the original coordinates the total potential is

$$V_t(x) + \frac{1}{2}\langle x, \Sigma_t^{-1}x \rangle,$$

and Σ_t^{-1} is precisely the additional Gaussian Hessian in α'_t . In (2), however,

$$\mu_t = \rho_t \gamma_C, \quad \log \rho_t = \text{constant} - V_t(M_t y).$$

The quadratic part is fixed in the invariant reference law and is absent from the score in (13). Also, the covariance-only term H_t has zero logarithmic expansion in the C^{-1} metric by (11). Hence transport expansion is controlled by the transformed interaction Hessian, with no additive Σ_t^{-1} term. This removes the Gaussian contribution to α_t in the precise sense relevant to Lipschitz transport.

Established heat-kernel specialization

Let $C = A^{-1}$ and

$$C_t = A^{-1}(I - e^{-tA}), \quad \Sigma_t = A^{-1}e^{-tA}.$$

Then $M_t = e^{-tA/2}$, $D_t = I/2$, and $H_t = 0$. Formula (2) becomes

$$\rho_t(y) = Z^{-1}e^{-V_t(e^{-tA/2}y)} = U_{t/2}\rho_0(y), \quad (14)$$

where U is the Ornstein–Uhlenbeck semigroup with invariant law $\gamma_{A^{-1}}$, and (13) becomes $v_t = -\frac{1}{2}\nabla \log \rho_t$. This is exactly the Langevin transport flow analyzed by Shenfeld; under the multiscale Bakry–Emery bound his Theorem 2.2 gives a Lipschitz limiting transport. Thus the arbitrary-path formula above agrees with the known construction in the standard schedule.

Why a scalar rescaling cannot be the general answer

Take $V_0 = 0$ and $C = \text{diag}(c_1, \dots, c_d)$ in the heat-kernel schedule. The covariance eigenvalues of ν_t are

$$c_i e^{-t/c_i}.$$

After any scalar blow-up $x \mapsto r_t x$, the ratio of the i th and j th eigenvalues remains

$$\frac{c_i}{c_j} \exp\{-t(1/c_i - 1/c_j)\}.$$

If $c_i \neq c_j$, this tends to 0 or $+\infty$. No scalar rescaling—in particular no scalar chosen only from the largest eigenvalue α'_t —can yield a finite, full-dimensional terminal law. The matrix whitening (1)–(2) is essential.

Verification, scope, and novelty

The identities (1), (4), the covariance telescoping law, (10)–(12), and the heat-kernel specialization were independently checked on random noncommuting positive-definite matrices by the included script. The proof itself is algebraic and does not rely on the computation.

Bounded searches used the exact source title, “rescaling of the Polchinski semigroup,” “blow up the Dirac mass,” arbitrary covariance decomposition, and Ornstein–Uhlenbeck/transport variants. Shenfeld’s arXiv:2205.01642v4 explicitly contains the heat-kernel/Langevin special case. Faris gives the classical link between Gaussian convolution plus scaling and Ornstein–Uhlenbeck semigroups. No source found stated the arbitrary two-parameter covariance-whitening conjugacy (3)–(9). Novelty confidence is moderate: the core covariance identity is elementary and may be folklore.

The result solves the rescaling and supplies its transport velocity. Quantitative Lipschitz constants for a completely arbitrary noncommuting covariance schedule still require bounds on the transformed interaction Hessian. This is an estimate for the resulting transport, not a missing piece of the rescaling.

Human-review recommendation. Check the direction and composition conventions in (6) and (9), the L^1 approximation argument in (7), and the continuity-equation sign in (13). Then compare the meaning of “remove” in the source’s discussion of α_t with the fixed-reference cancellation following (13).

References

1. J. Serres, *Behavior of the Poincaré constant along the Polchinski renormalization flow*, arXiv:2208.08186; Commun. Contemp. Math. 26 (2024), Paper 2350035.
2. Y. Shenfeld, *Exact renormalization groups and transportation of measures*, arXiv:2205.01642v4 (2026), especially Section 2 and Theorem 2.2.
3. W. G. Faris, *Ornstein–Uhlenbeck and renormalization semigroups*, Moscow Math. J. 1 (2001), 389–405.